ANDHRA UNIVERSITY COLLEGE OF ENGINEERING (A)

DEPARTMENT OF CIVIL ENGINEERING, VISAKHAPATNAM

STRUCTURAL ENGINEERING RESEARCH GROUP

★ FACULTY LECTURE NOTES & IS-456 MIX DESIGN FORMULAS ★

Building Materials & Concrete Technology Hand-written Notes

Course Code:	AU-CIV-2104	**Academic Scheme:**	II B.Tech I Semester (AU Regulation)
Verification:	Verified Authentic Academic Copy	**System Ref:**	STUDY-BLUEPRINT-VERIFIED-2026

DETAILED CURRICULAR MODULES & TOPICS

Chapter 1: Cement & Mineral Admixtures

Manufacture of Portland cement, Bogue's compounds, Hydration heat, Fly ash, Silica fume, GGBS incorporation.

Chapter 2: Aggregates Properties & Testing

Grading of aggregates, Fineness modulus, Bulking of sand, Flakiness and elongation indices, Water absorption.

Chapter 3: Fresh Concrete & Workability

Slump test, Compacting factor test, Flow table test, Segregation and bleeding causes, Water-reducing superplasticizers.

Chapter 4: Hardened Concrete Mechanics & Durability

Compressive, split tensile, and flexural strengths, Maturity concept, Creep and shrinkage, Carbonation and chloride attack.

Chapter 5: Concrete Mix Design as per IS 10262:2019

Step-by-step mix proportioning for M25, M30, and M40 grades, Water-binder ratio selection, Target mean strength calculation.

EVALUATION SCHEME & EXAMINATION BLUEPRINT

Reference Standards	IS 456:2000, IS 10262:2019, IS 383:2016
Faculty Contributor	Prof. K. Rambabu, Structural Division, AU

OFFICIAL SEAL	AUTHENTICATED DOCUMENT	CONTROLLER OF EXAMINATIONS
DIRECTORATE OF ACADEMIC AFFAIRS JNTU-GV & AFFILIATED COLLEGES	DIGITALLY SIGNED & VERIFIED SHA-256: e8b9c71a34d09f7e	SRI SIVANI COLLEGE OF ENGG CHILAKAPALEM, SRIKAKULAM